

The 24–31 gap: Computational methods and structural barriers for minimal Kochen–Specker sets in dimension three

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The minimum number of rays required for a Kochen–Specker set in dimension 3 is bounded between 24 (established 2023) and 31 (circa 1990), a gap whose current width has persisted since the SAT-based lower-bound improvements of Li–Bright–Ganesh and Kirchweiger–Peitl–Szeider. We report a systematic search for KS sets with fewer than 31 vectors using seven complementary strategies: SAT-verified minimization within six algebraic ray pools, cross-pool mixing, expanded coordinate alphabets (30 tested, including Pisot-number fields), numerical optimization, deletion-criticality analysis, triad density thresholds, and a graph perturbation search over 500,000 trials. No sub-31 KS set is found by any strategy. An OCUS procedure certifies that 31 is the exact minimum within the 49-ray integer pool $\{0, \pm 1, \pm 2\}^3$. The graph perturbation search finds 61,702 abstract KS-uncolorable hypergraphs with fewer than 31 vertices; numerical realizability testing finds no $\mathbb{R}^3$ embedding for any of them (best residual 3.8×10^{-2}). A complementary vertex-merging analysis produces 394 combinatorially valid 30-vertex KS graphs, all proved unrealizable within the integer pool by finite SAT encoding. The convergence of exact and numerical evidence identifies a *realizability barrier*: abstract sub-31 KS candidates exist in abundance, but none has been realized as rays in $\mathbb{R}^3$. We establish an empirical *modulus-2 boundary*—among integer-like alphabets $\{0, \pm 1, \pm x\}$, only those where $|x|^2 = 2$ (or the degenerate case $x = 2$) produce KS-uncolorable pools—and analyze structural constraints (constraint budget, plane matching property) relevant to proof attempts. The paper surveys prior work on KS minimality, taxonomizes the available methods of attack, and maps the landscape of approaches for closing the 24–31 gap.

I. INTRODUCTION

The Kochen–Specker (KS) theorem [1] (see Ref. [15] for a comprehensive review) establishes that quantum measurements in Hilbert spaces of dimension $d \geq 3$ cannot be described by noncontextual hidden-variable theories. Formally, the theorem is witnessed by a finite set of rays in $\mathbb{R}^d$ (or $\mathbb{C}^d$; equivalently, unit vectors up to scalar multiples) together with a collection of mutually orthogonal triples—the *bases* or *contexts*—such that no assignment of values in $\{0, 1\}$ to the rays can simultaneously satisfy two conditions: (i) exactly one ray in each basis receives the value 1, and (ii) if two rays from distinct bases are collinear, they receive the same value. Any such configuration is called a *KS set*. Its existence rules out hidden-variable completions that assign pre-measurement values independently of context.

Determining the minimum number of rays required to form a KS set in dimension 3 is one of the central open problems in quantum foundations. Despite more than five decades of effort, a definitive answer remains out of reach. The current situation is captured by the inequality $24 \leq n_{\min}^{(3)} \leq 31$, and closing this gap is the motivating problem of the present paper.

This paper is a companion to Ref. [20], which establishes the algebraic framework—six discrete “algebraic islands” of KS-supporting number rings, two cancellation mechanisms, and the pool construction methodology—on which the present search is built. We assume that framework here and do not re-derive it; the reader is referred to Ref. [20] for definitions, proofs, and the full classification. The present paper’s contribution is the system-

atic search for sub-31 KS sets using that framework, the identification of structural barriers, and the mapping of approaches for closing the gap.

History of the upper bound

The first explicit KS proof in dimension 3, due to Kochen and Specker [1], used 117 rays. Over the following decades, that number fell steadily. Peres [2] provided a 33-vector proof drawing rays from the coordinate alphabet $\{0, \pm 1, \pm \sqrt{2}\}$, and later reported the Conway–Kochen 31-vector set (CK-31) in his textbook [3, 4]. The CK-31 construction works entirely within the integer alphabet $\{0, \pm 1, \pm 2\}$: exact orthogonalities arise from the cancellation identity $1^2 + 1^2 = (\sqrt{2})^2$, which in integer coordinates reads $1 + 1 = 2$ after squaring. This arithmetic mechanism produces a remarkably compact combinatorial structure—31 rays arranged in 17 orthogonal triples (bases)—that has resisted all attempts at reduction for over three decades. The CK-31 set therefore stands as the smallest *known* KS set in dimension 3.

Cabello’s recent construction [18], using complex coordinates drawn from the Eisenstein integers $\mathbb{Z}[\omega]$ with $\omega = e^{2\pi i/3}$, yields 33 rays and provides an independent upper bound at 33, distinguished by minimizing the number of bases (14) rather than the number of rays. While it does not improve the upper bound on $n_{\min}^{(3)}$, it demonstrates that algebraic fields beyond $\mathbb{Z}$ can support small KS sets and illuminates the role of complex structure in KS geometry.

History of the lower bound

Progress on the lower bound has been slower but accelerating. Arends, Ouaknine, and Wampler [9] established a lower bound of 18 via systematic combinatorial search. Uijlen and Westerbaan [11] raised this to 22 using a SAT-based approach with certified uncolorability. The bound advanced to 24 independently and concurrently: Li, Bright, and Ganesh [17] used SAT solvers augmented with algebraic realizability checking, while Kirchweber, Peitl, and Szeider [16] employed co-certificate learning with SAT modulo symmetries. Both works confirm that no KS set on fewer than 24 rays exists in $\mathbb{R}^3$ or $\mathbb{C}^3$. The gap between 24 and 31 is the current frontier.

Rigidity and the conjecture that 31 is optimal

Trandaafir and Cabello [19] recently proved that CK-31 is *rigid*: it is unique up to unitary equivalence in $\mathbb{C}^3$, and every sub-configuration of fewer than 31 rays drawn from its basis structure is KS-colorable. They further conjectured, subject to two assumptions—that the minimum KS set is rigid and that it contains the minimal state-independent contextuality (SI-C) set—that $n_{\min}^{(3)} = 31$. This conjecture represents the sharpest current theoretical statement about the problem and provides the primary motivation for the present investigation: do the methods available today support or refute it?

Our contribution

We report a systematic, multi-strategy computational search for KS sets with fewer than 31 vectors. Seven complementary methods are deployed, spanning algebraic enumeration, SAT-based optimization, numerical gradient descent, graph-theoretic perturbation, and structural analysis. No sub-31 KS set is found by any strategy.

Our main findings are as follows. First, within each of six algebraic ray pools identified in our companion paper [20], MUS (minimum unsatisfiable subset) extraction consistently finds sharp minima: 31 rays for the integer pool, 33 for Peres, Eisenstein, and $\mathbb{Z}[\sqrt{-2}]$, 43 for Heegner-7, and 52 for the golden ratio. For the integer pool, an OCUS (optimal constrained unsatisfiable subset) procedure exhaustively certifies that no KS-uncolorable subset of ≤ 30 rays exists among all 49 pool rays; for the remaining pools, the minima are best-known values supported by thousands of independent MUS extractions but not exhaustively certified.

Second, we identify an empirical *modulus-2 boundary*: among integer-like alphabets of the form $\{0, \pm 1, \pm x\}$, only those where $|x|^2 = 2$ produce KS-uncolorable pools. Alphabets with $|x|^2 \geq 3$ (such as $\{0, \pm 1, \pm 3\}$) are colorable despite generating orthogonal triples. In a companion paper [20], we established that KS-uncolorability

in dimension 3 requires one of two cancellation mechanisms: modulus-2 cancellation ($|x|^2 = 2$, of which the integer identity $1 + 1 = 2$ is the degenerate case) or phase cancellation (a vanishing sum of roots of unity, as in the Eisenstein field). The modulus-2 boundary reported here is the integer-alphabet manifestation of this broader pattern.

Third, a graph perturbation search over 500,000 random trials finds 61,702 abstract KS-uncolorable hypergraphs with fewer than 31 vertices; numerical optimization finds no $\mathbb{R}^3$ realization for any of them (best residual 3.8×10^{-2} , six orders of magnitude above the acceptance threshold). A complementary vertex-merging analysis produces 394 distinct 30-vertex KS-uncolorable graphs derived from CK-31; all 394 are proved unrealizable within the integer pool by finite SAT encoding (< 1 s each). The convergence of these two independent lines of evidence—one numerical, one exact within a finite pool—identifies a *realizability barrier*: abstract sub-31 KS candidates exist in abundance, but embedding them as rays in $\mathbb{R}^3$ appears to be the core obstruction.

Fourth, we analyze the structural reasons why naive counting and coloring arguments are insufficient to close the gap, and identify the *plane matching property*—the requirement that every ray participate in sufficiently many orthogonal triples drawn from a fixed geodesically constrained configuration—as a candidate geometric constraint for future proof attempts.

Organization

Section II surveys prior work on KS minimality in detail. Section III introduces our computational framework, including the SAT encoding, realizability oracle, and software stack. Sections IV–IX present six search strategies: exact minimization within algebraic pools, cross-pool mixing, expanded alphabets, numerical optimization, criticality analysis, and triad density thresholds. Section X presents the graph perturbation search. Section XI analyzes vertex merging and the realizability barrier. Section XII develops structural insights toward a proof of optimality. Sections XIII and XIV discuss the landscape of approaches and state the open problems that remain.

The paper is written as an exploration of what current methods can and cannot achieve, addressed to researchers in quantum foundations, combinatorics, and automated reasoning. The central question—whether $n_{\min}^{(3)} = 31$ —remains open, but the evidence assembled here points consistently in one direction.

II. PRIOR WORK ON KS MINIMALITY

We organize the literature around four themes: the succession of upper bounds, the progression of lower

bounds, recent rigidity results, and the algebraic framework that underlies our search.

A. Upper bounds

The first explicit proof of the Kochen–Specker theorem in dimension 3 was given by Kochen and Specker themselves in 1967 [1]. Their construction used 117 rays arranged in a set of interlocking orthogonal triples that admits no consistent $\{0, 1\}$ -coloring. While the proof was conceptually foundational, the sheer size of the configuration left open the question of how compact a KS proof could be made.

The first significant reduction came from Peres [2], who in 1991 exhibited two 33-vector proofs drawing coordinates from the alphabet $\{0, \pm 1, \pm \sqrt{2}\}$. The non-integer entry $\sqrt{2}$ was essential: Peres exploited the identity $1^2 + 1^2 = (\sqrt{2})^2$ to produce exact orthogonalities that cannot be achieved with integer coordinates alone. In the same work, Peres noted that John H. Conway and Simon Kochen had communicated, around 1990, a 33-vector construction using only integer coordinates (subsequently referred to as CK-33). Both the Peres 33-vector set and CK-33 contain 33 vectors; they differ in their alphabets and in their orthogonality graphs.

Conway and Kochen subsequently improved their construction to 31 vectors, using coordinates drawn exclusively from the integer alphabet $\{0, \pm 1, \pm 2\}$. This CK-31 construction was reported by Peres in his 1993 textbook [3, 4]. The cancellation underlying CK-31 is the identity $1 + 1 = 2$ (equivalently, $1^2 + 1^2 + (-1)(2) = 0$ in a suitable triple product), which produces 17 mutually orthogonal triples from just 31 rays. CK-31 has remained the record holder for over three decades; no subsequent work has found a KS set in dimension 3 with fewer than 31 rays.

A notable observation about all upper bounds in dimension 3 is their reliance on algebraic coordinate structure [8, 10]. Every known KS set in $\mathbb{R}^3$ with ≤ 33 rays has coordinates drawn from a discrete, algebraically defined alphabet: the integers $\{0, \pm 1, \pm 2\}$, the ring $\mathbb{Z}[\sqrt{2}]$, or a cyclotomic extension. No “generic” or “numerically generic” KS set with an irregular coordinate distribution has ever been found. This empirical rigidity of the upper bound motivates the algebraic approach taken in the present paper.

Cabello [18] recently introduced a 33-vector KS set constructed from vectors over the Eisenstein integers $\mathbb{Z}[\omega]$ with $\omega = e^{2\pi i/3}$, using a Weyl–Heisenberg symmetry to organize the construction. This set uses only 14 bases (orthogonal triples), fewer than any previously known KS set in dimension 3, and was proposed as the “simplest” KS set by the criterion of minimizing the number of measurement contexts. It does not improve the upper bound on the number of rays ($33 > 31$), but it establishes that fields beyond the integers can produce small KS sets and that the number of bases is a distinct minimization

target.

For completeness, we mention record constructions in dimensions $d > 3$, which demonstrate that the minimum number of rays required for a KS proof decreases sharply with dimension. Kernaghan [5] found a 20-vector KS set in dimension 4, and Kernaghan and Peres [6] gave a 36-vector construction in dimension 8. Cabello, Estebaranz, and García-Alcaine [7] subsequently reduced the dimension-4 record to 18 vectors. These results are not directly comparable to the dimension-3 problem: the combinatorial and algebraic constraints differ fundamentally between dimensions, and a KS set for $d = 4$ does not embed into $\mathbb{R}^3$.

B. Lower bounds

Establishing that any KS proof in dimension 3 requires at least n rays is significantly harder than exhibiting a KS set. Early lower bounds were modest: simple counting and coloring arguments show that a KS set in dimension 3 needs at least 5 rays, and incremental combinatorial improvements over several decades brought this to 15.

Arends, Ouaknine, and Wampler [9] raised the lower bound to 18 in 2011 using a systematic approach combining exhaustive enumeration of candidate orthogonality hypergraphs with SAT-based coloring checks. Their method established that every 3-uniform hypergraph on fewer than 18 vertices whose hyperedges form a system of mutually orthogonal triples admits a valid $\{0, 1\}$ KS-coloring in $\mathbb{R}^3$.

Uijlen and Westerbaan [11] improved the bound to 22 in 2016 using stronger combinatorial constraints on the orthogonality graph. Their approach incorporated more refined structure on the allowed hypergraph types, ruling out additional families of small candidate configurations.

Two independent groups then established the current best lower bound of 24 vectors.

Li, Bright, and Ganesh [17], in work presented at IJCAI 2024, developed a pipeline that works in three stages: (1) enumerate abstract 3-uniform hypergraphs on n vertices that are KS-uncolorable (as verified by a SAT solver); (2) filter these by checking, again with a SAT solver, that the abstract uncolorability cannot be destroyed by any assignment of triples; and (3) test whether each surviving abstract hypergraph can be *geometrically realized* as actual rays in $\mathbb{R}^3$ by submitting the embeddability query to Z3’s nonlinear arithmetic solver. The geometric realizability step is the computational bottleneck: Z3 returns “unknown” on embeddability queries for hypergraphs of order greater than approximately 20, because the resulting systems of polynomial equations exceed the solver’s decision capabilities. Despite this bottleneck, the pipeline is rigorous: any abstract hypergraph for which Z3 certifies non-realizability is definitively ruled out. Li, Bright, and Ganesh concluded that no KS set on fewer than 24 rays exists in $\mathbb{R}^3$.

Kirchweger, Peitl, and Szeider [16], in independent

work presented at AAAI 2023, obtained the same lower bound of 24 using a different technique: co-certificate learning within a SAT modulo symmetries framework. Their approach generates symmetry-breaking clauses automatically to prevent the enumeration from revisiting isomorphic configurations, and uses co-certificates (proofs of unsatisfiability) to certify that each explored configuration is genuinely unrealizable. The method does not rely on external solvers for realizability and provides a self-contained proof.

Both groups confirm that no KS set exists on fewer than 24 rays in $\mathbb{R}^3$ or $\mathbb{C}^3$. Neither group claims that 24 is tight—the actual minimum $n_{\min}^{(3)}$ may lie anywhere in the interval [24, 31]. Closing this gap remains the central open problem in KS minimality, and the primary challenge is the realizability bottleneck: abstract KS-uncolorable hypergraphs with far fewer than 31 vertices exist in abundance, but verifying whether any can be embedded as real or complex rays requires algebraic tools that currently do not scale to the problem.

C. Rigidity and uniqueness

A KS set is *rigid* if every other set of rank-1 projectors in $\mathbb{C}^3$ whose orthogonality graph is isomorphic to that of the original must be unitarily equivalent to it. Rigidity is a strong uniqueness statement: it implies that the geometric realization is essentially unique, up to the symmetries of Hilbert space.

Trandafir and Cabello [19] recently proved that CK-31 is rigid in $\mathbb{C}^3$. Specifically, they showed that any collection of 31 rank-1 projectors satisfying the same orthogonality relations as CK-31—i.e., any geometric realization of the CK-31 orthogonality graph—must be unitarily equivalent to the original Conway–Kochen construction. This means CK-31 admits no non-trivial deformations: it cannot be continuously or discretely modified while preserving its combinatorial structure. In the same work, they proved the analogous rigidity result for CK-33 (the earlier Conway–Kochen 33-vector set, which is distinct from both CK-31 and the Peres 33-vector construction).

The rigidity result has a direct implication for the minimality problem. Because CK-31 is rigid, any KS set with fewer than 31 rays must have an orthogonality graph that is *not isomorphic* to that of CK-31. It cannot arise by a minor modification, perturbation, or deformation of the existing 31-ray construction: a sub-31 KS set, if it exists, would require an entirely different combinatorial structure.

Trandafir and Cabello also formulated a conditional conjecture: subject to the assumptions that the minimum KS set is rigid and contains the minimal SI-C set, they conjectured that $n_{\min}^{(3)} = 31$. This conjecture is the sharpest theoretical statement currently available about KS minimality in dimension 3 and provides the principal motivation for the computational investigation reported in the present paper.

D. The algebraic framework

All known small KS sets in dimension 3 exploit specific algebraic structures in their coordinate systems. Exact orthogonality between two integer-coordinate rays requires a vanishing inner product over $\mathbb{Z}$, which in turn requires the existence of a *cancellation identity*: an algebraic relation among coordinate values that allows their pairwise products to sum to zero. The abundance or scarcity of such relations determines how many orthogonal triples a given coordinate alphabet can support, and therefore how small a KS set constructed from that alphabet can be.

In a companion paper [20], we carry out a systematic survey of coordinate alphabets arising from quadratic, cyclotomic, and golden-ratio number fields, identifying which fields support KS constructions in dimension 3 and characterizing the algebraic mechanism in each case.

The survey identifies six *discrete algebraic islands*: disjoint families of coordinate alphabets, each internally consistent and each supporting a distinct smallest KS size. The islands correspond to (i) the rational integers $\mathbb{Z}$ with alphabet $\{0, \pm 1, \pm 2\}$ (minimum 31 rays), (ii) the Eisenstein integers $\mathbb{Z}[\omega]$ (minimum 33), (iii) the ring $\mathbb{Z}[\sqrt{2}]$ associated with Peres (minimum 33), (iv) the Gaussian-like ring $\mathbb{Z}[\sqrt{-2}]$ (minimum 33), (v) the Heegner-7 ring $\mathbb{Z}[\frac{1+\sqrt{-7}}{2}]$ (minimum 43), and (vi) the golden-ratio ring $\mathbb{Z}[\varphi]$ with $\varphi = \frac{1+\sqrt{5}}{2}$ (minimum 52). These six families are the only algebraic sources of KS constructions in dimension 3 found within the search space.

Two distinct cancellation mechanisms account for all observed KS-uncolorability. The first is *modulus-2 cancellation*: the generating algebraic integer x satisfies $|x|^2 = 2$, allowing the inner-product identity $x\bar{x} = 2$ to produce exact orthogonal triples. The pure integer case, $1 + 1 = 2$, is the degenerate additive instance of this mechanism. The second is *phase cancellation*: a vanishing sum of roots of unity, of which the simplest example is $1 + \omega + \omega^2 = 0$ for $\omega = e^{2\pi i/3}$, underlies the Eisenstein island. All six algebraic islands fall under one of these two mechanisms. Coordinate alphabets in which neither mechanism is available—including integer alphabets such as $\{0, \pm 1, \pm 3\}$ that lack the entry ± 2 —produce only colorable ray pools in all tested cases.

The companion paper also introduces two new KS constructions not previously reported in the literature: a 43-vector set based on the Heegner-7 ring and a 52-vector set based on the golden-ratio ring. These new constructions confirm the existence of the corresponding algebraic islands and demonstrate that KS geometry is richer than a focus on integer coordinates alone would suggest.

The present paper builds directly on this algebraic framework. Rather than searching over arbitrary ray configurations, we restrict attention to the six identified algebraic pools and ask whether any of them supports a KS set with fewer than 31 vectors, using the algebraic

TABLE I. The six algebraic ray pools. See Ref. [20] for construction details.

Pool	Ring	Rays	Triads
Integer	$\mathbb{Z}$	49	26
Eisenstein	$\mathbb{Z}[\omega]$	57	22
Peres	$\mathbb{Z}[\sqrt{2}]$	49	16
$\mathbb{Z}[\sqrt{-2}]$	$\mathbb{Z}[\sqrt{-2}]$	49	16
Heegner-7	$\mathbb{Z}[\alpha]$	145	42
Golden ratio	$\mathbb{Z}[\varphi]$	205	166

structure to make the search tractable.

III. FRAMEWORK AND METHODS

A. Algebraic pool construction

A *coordinate alphabet* $\mathcal{A}$ is a finite set of values from which ray coordinates are drawn. The *induced ray pool* $S(\mathcal{A})$ consists of all projectively distinct nonzero vectors in $\mathcal{A}^3$. Rays are canonicalized: the first nonzero coordinate is taken positive, integer coordinates are gcd-reduced to primitive form, and complex coordinates are placed in a standard lexicographic normal form. The *orthogonality graph* of a pool has rays as vertices and edges between orthogonal pairs (dot product zero); *triads* are 3-cliques in this graph—mutually orthogonal triples that form a basis up to normalization. See Ref. [20] for the full algebraic treatment of pool construction, canonicalization, and the ring-theoretic conditions on $\mathcal{A}$ that determine whether $S(\mathcal{A})$ contains any triads.

The six algebraic pools studied in this series are summarized in Table I. Each arises from a distinct number ring, and the pool sizes and triad counts determine the difficulty of the associated SAT and optimization problems.

B. SAT encoding of KS-uncolorability

KS-uncolorability is encoded as a propositional satisfiability (SAT) instance as follows. Introduce a Boolean variable b_v for each ray v in the pool, with the intended interpretation that $b_v = 1$ means v is colored “green” (assigned value 1). For each triad $\{a, b, c\}$, enforce an exactly-one constraint via the at-least-one clause

$$(b_a \vee b_b \vee b_c) \quad (1)$$

together with three pairwise-exclusion clauses $(\neg b_\alpha \vee \neg b_\beta)$ for each pair $\alpha \neq \beta \in \{a, b, c\}$. For each orthogonal pair (u, v) not sharing any triad in the pool, add the exclusion clause $(\neg b_u \vee \neg b_v)$. The resulting propositional formula is unsatisfiable if and only if no valid KS coloring of the ray set exists, i.e., the set is a KS set. This encoding is the standard extension of the triad constraint to non-basis

orthogonal pairs, and is strictly stronger than requiring only the triad constraints. All SAT solving uses the Glucose4 solver [12] via the PySAT interface.

C. MUS and OCUS extraction

A *Minimal Unsatisfiable Subset* (MUS) of a propositional formula is a subset of clauses that is itself unsatisfiable but becomes satisfiable upon removal of any single clause. In the KS encoding, a MUS identifies a minimal subset of orthogonality and triad constraints that already forces uncolorability. To extract MUSes at the *ray level*—identifying minimal KS-uncolorable ray subsets—we introduce a selector literal a_r for each ray r , and condition all clauses involving r on the assumption $a_r = \text{true}$. The solver then produces an unsatisfiable core over selector literals. This core is minimized by iterative deletion: we remove one selector at a time and check whether the residual formula remains unsatisfiable; a selector is discarded if its removal preserves unsatisfiability, retained otherwise. After minimization, the active selectors identify a minimal KS-uncolorable ray subset. Each independent MUS extraction (with randomized deletion order) may produce a different minimal subset, enabling enumeration of distinct minimal KS sets within a pool.

OCUS (Optimal Constrained Unsatisfiable Subset) strengthens MUS extraction by certifying the *globally minimum* unsatisfiable subset size within the pool. OCUS iteratively proves, for increasing $k = 1, 2, \dots$, that no unsatisfiable subset of $\leq k$ rays exists, until an unsatisfiable subset of size k^* is found and its minimality is certified. This provides an exhaustive lower bound on the minimum KS subset size within a finite pool, at the cost of substantially greater computational effort than single-MUS extraction.

D. Cross-product completion

For some algebraic fields, the raw alphabet $\mathcal{A}$ produces a ray pool with too few triads for KS-uncolorability. *Cross-product completion* expands the pool by adding rays of the form $v_i \times v_j$ for each orthogonal pair (v_i, v_j) already in the pool, where $\times$ denotes the standard vector cross product in $\mathbb{R}^3$. The cross product of two vectors whose coordinates lie in a number field K also has coordinates in K , so the expanded pool remains within the same algebraic family. The closure step is iterated until no new rays are generated. This procedure is essential for the golden-ratio pool: the raw alphabet $\{0, \pm 1, \pm \varphi\}$ produces too few triads to be KS-uncolorable, while the cross-product-completed pool of 205 rays with 166 triads contains a 52-ray KS island that is invisible without this closure step. See Ref. [20] for the full procedure and its ring-theoretic justification.

E. Realizability testing

Combinatorial KS-uncolorability (established by SAT) is a necessary but not sufficient condition for a ray set to exist as an actual physical configuration in $\mathbb{R}^3$: the abstract orthogonality graph must additionally be *realizable*, meaning there must exist actual unit vectors in $\mathbb{R}^3$ whose pairwise dot products match the graph’s edge structure. We employ three complementary methods for realizability testing, each with distinct strengths and limitations.

Z3/SMT. Encode orthogonality as polynomial equations over the reals: for each required-orthogonal pair (v_i, v_j) , impose $\sum_k v_{i,k} v_{j,k} = 0$, subject to the normalization constraints $\|v_i\| = 1$. The Z3 SMT solver [13] can decide satisfiability of such systems exactly using quantifier elimination over nonlinear real arithmetic (NRA). This approach is exact and correct, but the NRA decision procedure is doubly exponential in the number of variables; in practice, Z3 returns “unknown” for instances with more than approximately 20 free variables, making it inapplicable to most of the candidate sets encountered in this work.

Numerical optimization. Minimize the orthogonality residual $\sum_{(i,j) \in E} (\langle v_i, v_j \rangle)^2$ subject to $\|v_i\| = 1$ for all i , using L-BFGS-B with analytically computed gradients and 10–20 random initializations. A final residual below 10^{-10} is taken as evidence of realizability. This approach is fast and scales to large instances, but provides only one-sided evidence: a small residual indicates realizability (up to numerical precision), while a large residual does not prove unrealizability, since the optimizer may have converged to a local minimum.

Finite pool SAT. For realizability *within a specific algebraic pool* of m candidate rays, introduce Boolean variables $x_{v,r}$ (vertex v of the abstract graph assigned to pool ray r), with three constraint families: (i) each vertex is assigned exactly one ray ($\sum_r x_{v,r} = 1$); (ii) distinct vertices receive distinct rays ($x_{v,r} + x_{w,r} \leq 1$ for $v \neq w$); and (iii) if vertices v and w are required to be orthogonal, their assigned rays must be orthogonal in the pool. This reduces realizability within the pool to pure Boolean satisfiability, solvable in under 1 s for the instances that defeat Z3. The trade-off is scope: the method tests realizability within a finite algebraic pool, not in $\mathbb{R}^3$ generally. A negative answer (UNSAT) means the abstract graph has no realization within that pool; it does not rule out realizations in other pools or over the reals.

IV. MINIMIZATION WITHIN ALGEBRAIC POOLS

For each of the six algebraic ray pools (Table I), we applied three heuristic minimization methods in concert with the SAT encoding of Section III: (a) clause-level MUS extraction with randomized deletion, (b) greedy ray deletion with 2000 random orderings plus determinis-

TABLE II. Minimum KS subset size per pool. “MUS min” is the minimum found across all MUS extractions. “Trials” is the number of independent MUS extractions performed. Greedy and swap searches confirmed these minima in every case.

Pool	Rays	Triads	MUS min	Trials
Integer	49	26	31	5000
Eisenstein	57	22	33	1000
Peres	49	16	33	1000
$\mathbb{Z}[\sqrt{-2}]$	49	16	33	1000
Heegner-7	145	42	43	1000
Golden ratio	205	166	52	1000

tic orderings (ascending and descending norm, ascending and descending degree), and (c) swap-based local search in which one ray at a time is replaced by a pool ray not currently in the set, with the swap accepted whenever KS-uncolorability is preserved. All three methods use Glucose4 as the underlying SAT oracle.

A. Per-pool minima

Table II summarizes the best result found per pool. For the integer pool, all 5000 independent MUS extractions returned a set of exactly 31 rays, and no greedy or swap trial ever produced a 30-ray KS-uncolorable subset. For the Eisenstein, Peres, and $\mathbb{Z}[\sqrt{-2}]$ pools, all 1000 MUS extractions per pool returned sets of exactly 33 rays. The Heegner-7 pool minimum is 43 and the golden-ratio pool minimum is 52.

Greedy deletion (2000 random orderings plus norm-sorted and degree-sorted deterministic orderings) and swap-based local search were run after each MUS extraction as a consistency check: in no case did either method locate a subset smaller than the MUS minimum. The convergence of three independent methods to the same minimum provides strong corroborating evidence that the reported values are true pool minima, though exhaustive proof is provided only by the OCUS procedure described below.

B. Structure of the six minimal 31-sets from the integer pool

The following structural analysis was first reported in our companion paper [20]; we summarize it here because the norm stratification and swap-isolation properties are directly relevant to the sub-31 search.

From 5000 independent MUS extractions on the integer pool, exactly six combinatorially distinct minimal 31-ray KS sets were recovered. (Two sets are considered distinct if they differ in at least one ray; we cannot rule out the existence of additional minimal 31-sets not reached by our randomized MUS extractions.) Together,

the six sets use 37 of the 49 available integer pool rays, leaving 12 rays unused by any minimal set.

The six sets exhibit sharp norm stratification with respect to $\|v\|^2 = v_1^2 + v_2^2 + v_3^2$:

- $\|v\|^2 \leq 3$ (the $\{0, \pm 1\}$ sub-alphabet skeleton, 13 rays): present in *all* six sets. These 13 rays form an invariant core.
- $\|v\|^2 = 5$ (12 rays): present in 5 of 6 sets.
- $\|v\|^2 = 6$ (12 rays): present in 4 of 6 sets.
- $\|v\|^2 = 9$ (highest-norm rays in the pool, 12 rays): present in *none* of the six sets.

The exclusion of all 12 highest-norm rays from every minimal 31-set is a striking structural feature: the $\{0, \pm 1, \pm 2\}$ alphabet places the highest norms at $\|v\|^2 = 9$ (e.g., $(2, 2, 1)$ type), and these rays are never needed once the lower-norm rays supply sufficient interlocking constraints. The invariant 13-ray core (all rays with $\|v\|^2 \leq 3$, i.e., rays using only $\{0, \pm 1\}$ coordinates) contributes 7 of the 26 integer-pool triads and forms a configuration that is itself colorable, but whose extension by a small number of norm-5 and norm-6 rays creates the interlocking necessary for uncolorability.

Pairwise comparison of the six minimal sets reveals that no two are connected by a single-ray swap: the minimum Hamming distance (number of ray replacements needed to transform one set into another) is 5 ray replacements. Despite this topological separation, all pairs share between 25 and 26 rays out of 31, yielding a Jaccard similarity of approximately 0.71. CK-31 is one of the six. All six share the same orthogonality graph (verified by VF2 isomorphism testing [20]), consistent with the rigidity result of Trandafir and Cabello [19].

C. OCUS exhaustive certification

Heuristic methods cannot certify the absence of sub-31 sets; for this we used the OCUS procedure. Starting from the full 49-ray integer pool and its SAT encoding, OCUS iteratively enumerates hitting sets of all MUSEs found so far, calling the SAT oracle to test each candidate hitting set for unsatisfiability. If a hitting set of size k is UNSAT, a k -ray KS set has been found; if no hitting set of size k is UNSAT, OCUS advances to size $k+1$ or certifies that no k -ray KS set exists within the pool.

Applied to the integer pool, OCUS exhausted all candidates of size ≤ 30 : the search completed in 272 iterations, requiring less than 0.2 seconds of wall-clock time on a single core. To provide a complete certificate across all sizes from the trivial lower bound upward, we also ran OCUS for all values of n from 21 to 30 individually; the full range was certified in a combined 0.5 seconds. The result is an exhaustive, proof-by-exhaustion certificate:

No KS-uncolorable subset of ≤ 30 rays exists within the full 49-ray integer pool $\{0, \pm 1, \pm 2\}^3$.

This rules out any sub-31 KS set constructible from the $\{0, \pm 1, \pm 2\}$ integer pool. Combined with the consistency across all six pools reported in Table II, we conclude: *no sub-31 KS set is constructible from any of the six known algebraic ray pools in isolation.*

V. CROSS-POOL MIXING

The per-pool minima of Section IV establish that no single algebraic pool yields a sub-31 KS set. However, this does not preclude the possibility that rays from distinct pools might combine to create a smaller KS set through *cross-pool orthogonalities*: orthogonal pairs where one ray belongs to pool A and the other to pool B , mediated by a shared algebraic sub-structure. This section tests that possibility.

A. Method

We formed all 15 pairwise pool combinations and the full union of all six pools. For each combination, we: (1) took the union of the two (or more) ray sets; (2) deduplicated rays by canonical form (first nonzero coordinate positive, gcd-reduced); (3) computed the full orthogonality graph of the combined ray set, including all cross-pool orthogonal pairs; and (4) ran 500 independent greedy minimization trials, each starting from a random ordering of the combined ray set and deleting rays greedily while preserving KS-uncolorability. For each combined pool, we recorded the minimum KS-uncolorable subset size found and the pool composition of the resulting minimal set.

B. Results

Table III summarizes results for selected combinations; all 15 pairwise results are consistent with the pattern described here. The 477-ray full union result is also included.

Despite significant cross-pool orthogonalities—for example, the Integer + Golden combination shares 78 cross-pool triads in which one or more rays from each pool participate together in an orthogonal triple—every minimized set draws its rays from a *single pool only*:

- Any combination that includes the Integer pool: minimum 31, achieved entirely by integer rays.
- Any combination that excludes the Integer pool: minimum 33, achieved entirely by rays from one non-integer pool.

TABLE III. Cross-pool mixing results for selected combinations. “Min” is the best result in 500 greedy trials. “Source” indicates the pool from which all rays in the minimal set are drawn. “Cross-triads” counts orthogonal triples involving rays from two or more pools.

Combination	Min	Cross-triads	Source
Integer + Eisenstein	31	14	Integer
Integer + Golden	31	78	Integer
Integer + Heegner-7	31	31	Integer
Eisenstein + Peres	33	8	Eisenstein
Peres + $\mathbb{Z}[\sqrt{-2}]$	33	12	Peres
Eisenstein + Golden	33	0	Eisenstein
All six pools (477 rays)	31	—	Integer

- The full 477-ray union of all six pools: minimum 31, achieved entirely by integer rays.

In each case, the greedy minimization procedure converged to a single-pool solution: cross-pool orthogonalities did not provide a pathway to a smaller mixed-pool KS set. The Integer pool’s minimum of 31 dominates all combinations in which it appears because its 31-ray sets already saturate the minimum achievable by any of the heuristic methods employed.

C. Caveat: heuristic limitations

Greedy deletion is biased toward large basins of attraction in the space of minimal KS sets. A mixed-pool minimal set with a small basin of attraction could in principle exist but be systematically missed by greedy methods. Exhaustive cross-pool search—testing all subsets of a combined pool for KS-uncolorability—is computationally infeasible for the larger combinations (e.g., the 477-ray union has 2^{477} subsets). The OCUS procedure of Section IV is in principle applicable to cross-pool combinations, but the integer pool OCUS already runs on a 49-ray universe; extending it to the 477-ray union would require a significantly larger search.

We conclude: *greedy cross-pool mixing finds no hybrid KS set smaller than the corresponding single-pool minimum, ruling out hybrid algebraic constructions from known fields subject to the heuristic nature of the search.*

VI. EXPANDED ALPHABETS

The six algebraic pools of Table I are each generated by a specific coordinate alphabet. A natural question is whether a *richer* alphabet—more coordinate values, or a larger generator bound in the same field—produces a denser orthogonality structure in which a smaller KS set can be found. This section reports a systematic survey of 30 expanded alphabets.

TABLE IV. Selected expanded alphabets. “KS” indicates that the pool is KS-uncolorable (— means colorable); “Best” is the smallest KS subset found by greedy minimization in 300–500 trials (— means not applicable).

Alphabet	Rays	Best	Note
$\{0, \pm 1, \pm 2\}$	49	31	Baseline (CK-31)
$\{0, \pm 1, \pm 2, \pm 3\}$	145	31	Absorbs to CK-31
$\{0, \pm 1, \pm 2, \pm 3, \pm 4\}$	289	36	Heuristic limit
$\{0, \pm 1, \pm 3\}$	49	—	Not KS
$\{0, \pm 1, \pm 4\}$	49	—	Not KS
Eisenstein $c=3, n \leq 6$	345	31	Absorbs to CK-31
Peres + ± 2	109	31	Absorbs to CK-31
Heegner-7 + ± 2	253	31	Absorbs to CK-31
Golden + ± 2	637	33	No absorption
$\{0, \pm 1, \pm 2, \pm i\}$	127	31	Mixed-field

A. Method

For each expanded alphabet, we: (1) enumerated all projectively distinct rays in $\mathbb{R}^3$ (or $\mathbb{C}^3$) whose coordinates lie in the extended alphabet; (2) computed the orthogonality graph and triad list; (3) verified KS-uncolorability of the full pool (some alphabets are not KS-uncolorable at all); and (4) ran 300–500 greedy minimization trials to find the smallest KS-uncolorable subset. Alphabets were drawn from four families: (i) extended integer alphabets $\{0, \pm 1, \dots, \pm k\}$; (ii) extended versions of each non-integer pool (larger generator bound or additional generator); (iii) mixed-field alphabets combining generators from two distinct rings; and (iv) Pisot-number fields (plastic ratio, tribonacci constant, silver ratio) not included among the original six pools.

B. Results

Table IV presents results for representative alphabets; the full 30-alphabet survey is consistent with the patterns described here.

C. The modulus-2 boundary

The most striking pattern in Table IV is a sharp boundary defined by the presence or absence of the coordinate value ± 2 .

Alphabets containing ± 2 converge to minimum 31. Every alphabet that contains the $\{0, \pm 1, \pm 2\}$ sub-alphabet as a subset converges, under greedy minimization, to a 31-ray KS set whose rays are drawn entirely from the integer pool:

- $\{0, \pm 1, \pm 2, \pm 3\}$ with 145 rays $\rightarrow$ minimum 31 (integer rays).
- $\{0, \pm 1, \pm 2, \pm i\}$ with 127 rays (mixed real-imaginary field) $\rightarrow$ minimum 31 (integer rays).

- Eisenstein $c = 3$, $n \leq 6$, with 345 rays $\rightarrow$ minimum 31 (integer rays).
- Peres $+ \pm 2$ (109 rays) and Heegner-7 $+ \pm 2$ (253 rays) $\rightarrow$ minimum 31 (integer rays).

In each case the addition of ± 2 to a non-integer pool causes greedy minimization to “absorb” the solution back to the CK-31 structure in the integer sub-pool. The 637-ray Golden $+ \pm 2$ pool is the sole exception: greedy minimization finds minimum 33, and the golden-ratio algebraic structure prevents absorption to the integer sub-pool.

Alphabets lacking ± 2 are not KS-uncolorable.

The alphabets $\{0, \pm 1, \pm 3\}$ and $\{0, \pm 1, \pm 4\}$ —which possess 49 rays and 10 triads each, comparable in size to the baseline integer pool—are *not KS-uncolorable at all*: the full pool is 2-colorable. Despite the availability of orthogonal triples, the triads cannot be interlocked in a way that forces contradiction. Analogously, $\{0, \pm 1, \pm 2, \pm 3, \pm 4\}$ with 289 rays reaches only minimum 36 under greedy minimization—higher than the integer pool minimum despite a much larger ray count—suggesting that the additional high-norm rays do not contribute usable orthogonal structure.

The mechanism is algebraic: the cancellation identity $1 + 1 = 2$ enables constructions such as $(1, 1, 1) \perp (1, 1, -2)$, providing a “two-against-one” triad type that is absent from alphabets like $\{0, \pm 1, \pm 3\}$. Without this identity, the only triads available are of “one-against-one” type or require larger coordinate magnitudes, and these do not produce a sufficiently dense interlocking structure for KS-uncolorability in dimension 3.

D. Pisot-number fields

We tested three higher-degree algebraic fields associated with Pisot numbers: the plastic ratio ($x^3 = x + 1$, degree 3), the tribonacci constant ($x^3 = x^2 + x + 1$, degree 3), and the silver ratio ($x^2 = 2x + 1$, degree 2). All three produce only colorable pools: despite generating orthogonal triples from their respective algebraic identities, no KS-uncolorable configuration was found. This extends the modulus-2 boundary observation beyond quadratic fields to degree-3 algebraic numbers: the Pisot-number cancellation identities are not sufficient for KS-uncolorability in the three-dimensional setting surveyed here.

E. Summary

Across all 30 tested alphabets, no sub-31 KS set was found. The results establish an empirical modulus-2 boundary for integer-like alphabets: among alphabets of the form $\{0, \pm 1, \pm x\}$, only those with $|x|^2 = 2$ produce KS-uncolorable pools. (Non-integer fields such as the Eisenstein integers use a different mechanism—phase

cancellation—and are not covered by this specific boundary; see Ref. [20] for the full two-mechanism classification.) We conclude: *expanded coordinate sets from all tested fields fail to produce sub-31 KS sets.*

VII. NUMERICAL OPTIMIZATION

Strategies 1–3 operate exclusively within finite algebraic ray pools: they can only find KS sets whose coordinates lie in the tested number rings. A natural question is whether continuous optimization, unconstrained by algebraic structure, can locate KS sets in parts of configuration space not accessible to algebraic methods. We tested four non-algebraic methods as a *negative control*.

Simulated annealing from scratch. We initialized random unit vectors in $\mathbb{R}^3$ (at sizes $n = 28, 29, 30$) and in $\mathbb{C}^3$ (at sizes $n = 28, 30$), then ran 2×10^5 annealing steps with an objective function that counts the number of exactly orthogonal pairs (inner product below a threshold $\epsilon = 10^{-8}$). Results: at most 5–6 exact orthogonal pairs were achieved in $\mathbb{R}^3$; zero exact orthogonal pairs were achieved in $\mathbb{C}^3$; and zero triads (mutually orthogonal triples) were found in either case across all tested sizes.

CK-31 perturbation. We removed $m = 1, 2, 3$ rays at random from CK-31, replaced them with random unit vectors drawn uniformly from S^2 , and tested whether the resulting set is KS-uncolorable. In 1000 independent attempts for each m , the best result achieved was preservation of 16 of CK-31’s 17 triads, but all 17 were never simultaneously preserved. KS-uncolorability was never restored.

Random orthogonal completion. Starting with 5 random seed rays, we iteratively extended the set by finding unit vectors orthogonal to existing pairs via the cross product $w = u \times v / \|u \times v\|$. Candidate sets of 60 rays achieved at most 1 triad. No KS-uncolorable set was found.

Soft-tolerance annealing. We allowed near-orthogonality (tolerance $\epsilon = 0.1$), then progressively tightened $\epsilon \rightarrow 0$. At tolerance 0.1, we found 227 near-orthogonal pairs per 30-ray configuration on average. However, as ϵ was tightened toward exact orthogonality, all near-orthogonal pairs evaporated: zero exact orthogonal pairs survived.

Total runtime across all four methods: approximately 12 minutes.

Key insight: exact orthogonality is measure-zero. In the space of unit vectors in $\mathbb{R}^3$ or $\mathbb{C}^3$, the condition $\langle u, v \rangle = 0$ defines a measure-zero set. For the unguided random-start methods tested here, there is no gradient signal connecting near-orthogonal configurations to exact ones: the “number of exact orthogonal pairs” objective is constant (zero) almost everywhere, jumping discontinuously on a set of measure zero. More sophisticated continuous methods (homotopy continuation, manifold-constrained search) are not ruled out in principle, but the

TABLE V. Extended criticality of CK-31. For $k \leq 8$, all $\binom{31}{k}$ subsets tested exhaustively; for $k = 9-12$, 10^6 random subsets sampled per k . All subsets colorable.

k removed	Remain	Subsets tested	Method
1–4	30–27	36,456	Exhaustive
5	26	169,911	Exhaustive
6	25	736,281	Exhaustive
7	24	2,629,575	Exhaustive
8	23	7,888,725	Exhaustive
9–12	22–19	4×10^6	Sampled

absence of any gradient signal in the tested landscape—and the complete evaporation of near-orthogonal structure under tolerance tightening—suggests that algebraic structure is necessary for producing the exact orthogonalities that KS sets require.

Interpretation. The failure of all four numerical methods is not independent evidence against the existence of a sub-31 KS set. A sub-31 set could exist with coordinates in an untested algebraic field. The negative control confirms that if such a set exists, it must have algebraic structure, and further search should focus on identifying algebraic fields with untested cancellation mechanisms.

VIII. CRITICALITY AND DELETION-MINIMALITY

A KS set S is *deletion-minimal* if removing any single ray from S produces a colorable set. Deletion-minimality implies that S has no proper KS subset: every proper subset is contained in some single-deletion subset, and every single-deletion subset is colorable; since colorability is hereditary (any subset of a colorable set is colorable), every proper subset is colorable.

CK-31 is deletion-minimal. We tested all 31 single-ray removals from CK-31 exhaustively via SAT. In each case, the resulting 30-ray set is colorable. This confirms that CK-31 contains no proper KS subset: *every* proper subset of CK-31 is colorable. This result was verified independently for all six distinct minimal 31-sets found in the integer pool; each is deletion-minimal.

Extended criticality testing. As a computational diagnostic, we performed extended criticality testing by removing k rays for $k = 1, \dots, 12$ and testing colorability of the remaining $31 - k$ rays. Results are summarized in Table V.

Total: 11,460,647 exhaustive SAT checks plus 4×10^6 sampled checks. *All subsets are colorable.*

Significance at the lower-bound threshold. At $k = 7$, the remaining 24 rays equal the Li–Bright–Ganesh lower bound [17]. All $\binom{31}{7} = 2,629,575$ subsets of CK-31 of size 24 are colorable, providing explicit confirmation at the lower-bound threshold. (This already follows from deletion-minimality, but the exhaustive check provides a

concrete certificate at exactly the boundary.)

Implication. The extended criticality data rules out all sub-configurations of CK-31 as potential smaller KS sets. Any sub-31 KS set must use rays *not* in CK-31—drawn from a different part of the integer pool, from a different algebraic pool entirely, or from an untested number ring. Together with the vertex-merging unrealizability result (Section XI), this closes off two natural routes to a sub-31 KS set within the integer pool: sub-configurations of CK-31 and combinatorial modifications of CK-31’s orthogonality graph.

IX. TRIAD DENSITY ANALYSIS

The previous sections establish negative results for specific constructions. A complementary question is: how rare are KS sets within their own pools? If KS sets are extremely rare at sizes just above the pool minimum, sub-threshold existence is unlikely, though density arguments cannot constitute a proof of optimality.

Method. For each pool, we sampled 500 uniform random subsets at each size n , tested each for KS-uncolorability via SAT, and recorded the fraction that are uncolorable. We use the rule-of-three to set upper bounds on prevalence when no uncolorable sets are observed: if k samples are drawn with no successes, the 95% upper bound is $p \lesssim 3/k$.

Integer pool (49 rays). No KS-uncolorable subset is observed below $n = 42$ out of 49 rays ($p \lesssim 0.006$ at 95% confidence). The gap between the SAT minimum of 31 and the density threshold of 42 reflects the extreme rarity of KS sets even within pools known to contain them.

Eisenstein pool (57 rays). No KS-uncolorable subset below $n = 48$ out of 57. The Eisenstein pool has a *higher* SAT minimum (33 vs. 31) despite $\mathbb{C}^3$ providing additional orthogonality freedom, demonstrating that working in $\mathbb{C}^3$ does not automatically improve the minimum size.

Heegner-7 pool (145 rays). No KS-uncolorable subset at any sampled size up to $n = 75$ out of 145. This pool has the largest gap between pool size and SAT minimum (43 vs. 145), consistent with its sparser orthogonality structure.

Pair density of CK-31. Among all tested constructions, CK-31 achieves the highest pair density: 15.3% of all $\binom{31}{2} = 465$ pairs are orthogonal.

Key insight: density alone does not determine uncolorability. KS-uncolorable sets are extraordinarily rare even within their own pools. The Eisenstein pool (complex coordinates) has a higher density threshold than the integer pool despite having additional orthogonality freedom. The boundary between colorable and uncolorable is determined by the *topological interlocking* of the constraint hypergraph, not by any simple numerical density measure.

Random rays in $\mathbb{R}^3$ or $\mathbb{C}^3$, drawn uniformly from the unit sphere without reference to any algebraic pool, pro-

duce zero orthogonal pairs at any tested size $n \leq 50$ across 500 samples, confirming from a different angle that algebraic structure is essential.

Implications for proof strategies. This analysis rules out density-based approaches to proving optimality. A proof that no sub-31 KS set exists cannot rely on counting orthogonal pairs or triads. The precise topological structure of the constraint hypergraph must be the object of study.

X. GRAPH PERTURBATION SEARCH

A. Motivation

Strategies 1–6 all search within or near established algebraic pools. If a sub-31 KS set exists with an orthogonality graph not arising from any known algebraic construction—one that is in some sense “irregular”—those strategies would miss it entirely. Strategy 4 (numerical optimization) attempts to escape algebraic pools, but exact orthogonality is measure-zero in continuous space, so random ray replacement quickly loses all triad structure. A more targeted approach is needed: one that searches the space of abstract orthogonality hypergraphs directly, without requiring coordinates, then tests realizability only for uncolorable candidates.

B. Method

We start from CK-31’s orthogonality graph: 31 vertices, 71 edges, 17 triads. Each trial applies between 1 and 5 perturbations drawn from five operations:

1. *Remove vertex* (weight 3): delete one vertex and all incident edges; remap remaining vertices.
2. *Swap triad* (weight 3): remove one triad and add a randomly chosen new 3-clique.
3. *Merge vertices* (weight 2): identify two non-adjacent vertices into one, combining their neighborhoods.
4. *Add pair* (weight 1): add an orthogonality edge between two non-adjacent vertices.
5. *Remove pair* (weight 1): remove an orthogonality edge not belonging to any triad.

After perturbation, we test KS-uncolorability via SAT (Glucose4). If UNSAT and $n < 31$, we test $\mathbb{R}^3$ realizability: minimize $\sum_{(i,j) \in E} (v_i \cdot v_j)^2$ for unit vectors $v_i \in \mathbb{R}^3$ using L-BFGS-B with analytical gradients and 10–15 random initializations, tolerance 10^{-10} .

TABLE VI. Graph perturbation search results (500,000 trials, 12.73 hours).

Metric	Count
Valid perturbations	500,000
Still KS-uncolorable	79,850 (16.0%)
Sub-31 and uncolorable	61,702 (12.3%)
Near-realizable (residual < 0.1)	1,904
Realizable (residual $< 10^{-10}$)	0
Best residual	3.8×10^{-2}
Best configuration	$n = 30, t = 17, p = 69$

C. Results

We ran 500,000 trials over 12.73 hours. Table VI summarizes the outcomes.

D. Interpretation

Abstract sub-31 KS-uncolorable hypergraphs are abundant. Roughly 12.3% of all valid perturbations produce abstract hypergraphs that have fewer than 31 vertices and are KS-uncolorable. The combinatorial requirement for a sub-31 KS set is not the bottleneck.

The geometric requirement is the bottleneck. Of the 61,702 abstract sub-31 candidates, numerical optimization (L-BFGS-B, 10–15 random starts per candidate) finds no $\mathbb{R}^3$ realization for any of them. The 1,904 near-realizable cases (residual < 0.1) represent the closest the optimizer reaches, and the best residual across all candidates is 3.8×10^{-2} —more than six orders of magnitude above the acceptance threshold of 10^{-10} . While numerical optimization cannot prove unrealizability, the systematic failure across 61,702 independent instances with consistently large residuals constitutes strong evidence of a structural obstruction rather than an optimization artifact.

The best residual signals a structural gap, not a near-miss. A residual of 3.8×10^{-2} reflects an incompatibility between the abstract graph’s structure and three-dimensional real geometry. This is qualitatively different from a near-miss where the residual is small but optimization is slow.

The best configuration closely mirrors CK-31. The configuration achieving the best residual has $n = 30$, $t = 17$, $p = 69$. CK-31 has $n = 31$, $t = 17$, $p = 71$. Same number of triads, two fewer pairs, one fewer vertex—but the geometry does not give way.

Remark 1. *The 61,702 abstract sub-31 KS-uncolorable hypergraphs are combinatorial objects: they exist as hypergraphs but no $\mathbb{R}^3$ realization has been found for any of them. Their existence shows that abstract KS-uncolorability is achievable below 31 vertices; the obstruction to a sub-31 KS set (i.e., a realized configuration of rays) appears to lie in the interaction of uncolorability requirements with three-dimensional Euclidean geometry.*

XI. VERTEX MERGING AND THE REALIZABILITY BARRIER

A. The vertex merging construction

We generate 30-vertex candidate KS graphs by *vertex merging*: identify two non-adjacent vertices u and v into a single vertex w whose neighborhood is $N(w) = N(u) \cup N(v) \setminus \{u, v\}$.

Proposition 1 (Merge preserves KS-uncolorability). *If G is the orthogonality graph of a KS-uncolorable ray set, and G' is obtained from G by merging two non-adjacent vertices u and v , then G' is KS-uncolorable.*

Proof. Suppose G' admits a KS coloring f' . Define $f : V(G) \rightarrow \{0, 1\}$ by $f(u) = f(v) = f'(w)$ and $f(x) = f'(x)$ for all other vertices. Since u and v are non-adjacent in G , pairwise exclusion is not imposed between them. Every triad and pair constraint in G involving at most one of $\{u, v\}$ is satisfied by f' . Constraints involving both u and v cannot arise from triads (since $u \not\sim v$). Thus f is a valid KS coloring of G , contradicting its uncolorability. $\square$

B. Applying merging to CK-31

CK-31 has $\binom{31}{2} - 71 = 394$ non-adjacent pairs. By Proposition 1, all 394 merges produce abstract 30-vertex KS-uncolorable graphs with 17–21 triads and 70–71 pairs.

C. Realizability testing: three approaches

Z3/SMT. We encoded $\mathbb{R}^3$ realizability as polynomial constraints and submitted each instance to Z3 [13]. Result: “unknown” on all 394. Intractable at this scale.

Finite pool SAT. Assign each of the 30 abstract vertices to one of the 49 integer pool rays, subject to injectivity, orthogonality preservation, and non-adjacency constraints. All 394 instances UNSAT in under 1 s each. Definitively rules out realizability within the $\{0, \pm 1, \pm 2\}$ pool.

Numerical optimization. L-BFGS-B with 10–15 random starts per instance. No realization found; best residuals range from 0.01 to 0.15.

D. The realizability barrier

Observation 1 (Realizability barrier). *Two independent lines of evidence converge:*

- (i) Vertex merging: *all 394 combinatorially valid 30-vertex KS graphs derived from CK-31 are proved unrealizable within the $\{0, \pm 1, \pm 2\}$ integer pool (finite pool SAT, all UNSAT in < 1 s each). This is an exact result within the pool; realizability in other pools or in $\mathbb{R}^3$ generally remains open.*

- (ii) Graph perturbation (Section X): *among 61,702 abstract sub-31 KS-uncolorable hypergraphs, numerical optimization finds no $\mathbb{R}^3$ realization (best residual 3.8×10^{-2}). This is strong numerical evidence but not a proof of unrealizability.*

In total, 62,096 abstract sub-31 KS-uncolorable hypergraphs have been tested (394 from merging, 61,702 from perturbation; overlap and isomorphism reduction not yet performed). No realization has been found by any method.

The evidence points to a *realizability gap*: abstract KS-uncolorable hypergraphs with $n < 31$ exist in abundance, but embedding them as rays in $\mathbb{R}^3$ appears to be the core obstruction. The 394 merged graphs are definitively excluded from the integer pool; the broader claim that no sub-31 abstract KS hypergraph is $\mathbb{R}^3$ -realizable is supported by extensive numerical evidence but remains unproved.

We distinguish this *structural* realizability barrier from the *computational* realizability bottleneck identified by Li, Bright, and Ganesh [17], whose pipeline stalls because Z3 returns “unknown” on embeddability queries above order ~ 20 . Their observation is that realizability *checking* is hard; ours is that realizability itself appears to *fail* for sub-31 candidates—62,096 tested, zero realized. The two findings are complementary: theirs explains why the lower bound has stalled at 24; ours suggests that closing the gap from above may require proving a geometric impossibility rather than finding a construction.

Remark 2. *The finite pool SAT encoding resolves realizability instances in under 1 s that defeat Z3. It is directly applicable as a “quick accept” screen in the Li–Bright–Ganesh pipeline [17], which is bottlenecked by Z3 above order 20.*

E. Open questions

The 394 merged graphs were tested only within the integer pool. Open: (1) realizability in other algebraic pools; (2) realizability in $\mathbb{C}^3$; (3) isomorphism classes among the 394 graphs; (4) vertex merging applied to other pools (e.g., 33-vector Eisenstein set).

XII. TOWARD A PROOF OF OPTIMALITY

The computational evidence is strong: no sub-31 KS set has been found by any of seven independent methods, and the realizability barrier (Observation 1) shows that abstract sub-31 KS hypergraphs are combinatorially plentiful but have resisted all attempts at geometric realization. We frame the following as structural insights and candidate proof directions, not as partial proofs.

A. Constraint budget

Suppose S is a minimal KS set in $\mathbb{R}^3$ with $|S| = n$. Let $t = |\mathcal{T}|$ denote the number of triads and $p = |E|$ the number of orthogonal pairs. Define $C = t + p$.

For CK-31: $n = 31$, $t = 17$, $p = 71$, $C = 88$, $C/n = 2.839$. The 71 pairs decompose into 51 triad-implied pairs (17×3) and 20 extra pairs not in any triad.

Observation 2 (Extra pairs are essential). *Every one of CK-31's 20 extra pairs is essential: removing any single extra pair makes the formula satisfiable. The 51 triad-implied pairs alone are satisfiable. CK-31 requires all 20 extra pairs simultaneously.*

CK-31 is *maximally tight*: its uncolorability is spread across the entire constraint structure with no slack.

B. Why counting arguments fail

Degree-1 vectors are essential. A naive intuition suggests every vector should participate in ≥ 2 triads. This is false: 18 of 31 CK-31 vectors participate in only one triad. These degree-1 vectors are essential via extra pair constraints, not triads.

Triads containing v	Count
1	18
2	8
3	3
4	2

High density does not imply uncolorability. The densest 30-vertex subsets of the integer pool achieve C/n up to 5.33—far above CK-31's 2.839—yet all are colorable. The precise topological interlocking, not density, determines uncolorability.

C. The plane matching property

Definition 1. *For a ray $v \in \mathbb{R}^3$, the perpendicular plane $v^\perp$ is 2-dimensional. A triad $\{v, u, w\}$ places u and w in $v^\perp$ as an orthogonal pair.*

Observation 3 (Plane matching property). *In $\mathbb{R}^3$, if v is contained in k triads, the $2k$ triad neighbors of v form k orthogonal pairs (a matching) within $v^\perp$. This is verified for all 31 vectors of CK-31.*

The non-triviality of this constraint lies in its consequences for abstract hypergraph realizability. In $\mathbb{R}^3$, the 2-dimensionality of $v^\perp$ means that a vertex of triad-degree k forces its $2k$ triad neighbors into k orthogonal pairs within a single plane—a matching constraint that severely limits the number of triads that can share a vertex (at most $\lfloor (n-1)/2 \rfloor$ triads through any single ray, and each imposes a rigid directional relationship on its

partners). In dimensions $d \geq 4$, $v^\perp$ has dimension ≥ 3 , orthogonal pairs within it need not form a matching, and the constraint vanishes. This dimensional restriction is a candidate explanation for why the minimum KS size in $\mathbb{R}^3$ is substantially larger than in $\mathbb{R}^4$ (where 18–20 vectors suffice).

The plane matching property is not itself a proof of optimality—it is an $\mathbb{R}^3$ -specific structural constraint that any realized KS set must satisfy. The open question is whether it, combined with other geometric constraints, suffices to exclude all abstract KS-uncolorable hypergraphs below 31 vertices.

Conjecture 1 (Local-global realizability gap). *There exist abstract KS-uncolorable hypergraphs on $n < 31$ vertices whose local triad structure at each vertex is individually compatible with the plane matching property in $\mathbb{R}^3$, but for which no globally consistent assignment of rays exists. That is, the obstruction to realizability is global, not detectable by local checks alone.*

D. What a proof would require

A proof of optimality cannot rely on counting arguments. It must connect $\mathbb{R}^3$ geometry (which constrains realizable orthogonality graphs via the plane matching property) with the combinatorial structure required for uncolorability.

The Li–Bright–Ganesh approach [17] enumerates abstract hypergraphs bottom-up; our approach identifies geometric constraints top-down. A proof likely needs both: LBG-style enumeration (possibly accelerated by our finite SAT encoding) plus geometric constraints (plane matching, rigidity).

Conjecture 2 (Optimality of CK-31). *No KS set in $\mathbb{R}^3$ has fewer than 31 vectors. Equivalently, every abstract KS-uncolorable hypergraph on at most 30 vertices is unrealizable in $\mathbb{R}^3$.*

Our computational evidence is consistent with this conjecture: 62,096 abstract sub-31 KS hypergraphs tested, zero realized.

XIII. DISCUSSION: THE LANDSCAPE OF APPROACHES

A. Taxonomy of approaches

The search for sub-31 KS sets has been pursued through four broad families of methods, each with characteristic strengths and limitations.

Algebraic search (Strategies 1–3 and 6; pool construction, alphabet expansion, and cross-pool mixing). The algebraic approach builds ray pools from coordinate alphabets drawn from specific number rings, then

applies SAT-based uncolorability testing and OCUS extraction to find or rule out KS sets within each pool. Its primary strength is exhaustiveness within a tested field: every subset of a finite pool is implicitly checked, and OCUS certification provides a formally verifiable guarantee that no smaller KS set exists in that pool. Results are SAT-certifiable and reproducible. The principal limitation is scope: there are infinitely many number fields, and untested fields may harbor novel cancellation identities that enable new KS constructions. Our contribution across this approach: six pools from distinct number rings are fully characterized; the modulus-2 boundary is identified as the algebraic obstruction separating 29-alphabet expansions (all failing) from 31-achieving constructions; 30 expanded alphabets are tested without yielding sub-31 sets; and OCUS provides a certified lower bound of 31 for the integer pool.

Combinatorial enumeration [16, 17] (abstract hypergraph generation followed by realizability checking). The combinatorial approach generates abstract KS-uncolorable hypergraphs directly, without reference to a specific coordinate field, and then asks whether each abstract graph can be realized by unit vectors in $\mathbb{R}^3$. Its theoretical strength is completeness: any KS set that exists will eventually appear as a node in the enumeration. The practical limitation is the realizability bottleneck. Above approximately order 20, the Z3 SMT solver returns “unknown” on the nonlinear real arithmetic queries that realizability requires, so the pipeline stalls even when abstract candidates are abundant. Our finite pool SAT encoding (Section III) resolves instances in under 1 s that defeat Z3, and is directly applicable as a “quick accept” screen in the LBG pipeline: a combinatorial candidate can be tested for membership in any of our six algebraic pools before an expensive Z3 query is invoked. The 61,702 abstract sub-31 candidates found in our perturbation search (Section X) provide a concrete input set for such screening.

Numerical and heuristic search (Strategies 4 and 7; simulated annealing, perturbation, and graph perturbation). Two additional approaches belong to this family but were not tested systematically: polytope-based generation from regular solids (tested for higher dimensions by Pavicic et al. [8] but less natural in $d = 3$ where no regular polytope produces KS sets), and homotopy continuation methods (noted by Pavicic et al. as having been “tested without success on representative systems”). Heuristic methods explore large search spaces without algebraic constraints, probing for KS sets that might lie outside any specific algebraic pool. Their strength is speed and breadth: 500,000 perturbation trials can be completed where an exhaustive algebraic search would be infeasible. The fundamental limitation is the absence of completeness guarantees: heuristic methods can find KS sets but cannot rule them out. Our contribution is the quantification of the realizability barrier. The perturbation search generated 61,702 distinct abstract KS-uncolorable graphs of order 28–30, yet none

was realizable in $\mathbb{R}^3$ under any of three complementary realizability tests. This systematic failure is not a null result: it establishes that the obstruction to sub-31 KS sets is not merely a lack of uncolorable graphs (which exist in abundance) but specifically the failure of those graphs to be embeddable in three-dimensional Euclidean space.

Structural and proof-theoretic methods (Section XII; Trandafir–Cabello rigidity [19]; constraint budget analysis; plane matching). Structural methods seek not merely to test whether sub-31 KS sets exist but to prove or disprove their existence via geometric or combinatorial arguments. This is the only approach capable of yielding a definitive answer to the central question. The limitation is that no such proof currently exists. Our contributions include: identifying why naive counting arguments fail to close the gap; formulating the plane matching property as a candidate geometric constraint that any sub-31 set would have to violate; and the constraint budget analysis showing that CK-31 is maximally tight in the sense that its orthogonality constraints leave no slack for simplification.

B. Connections to published work

Trandafir–Cabello rigidity [19]. Trandafir and Cabello proved that CK-31 is unique up to unitary equivalence: any KS set with exactly the same orthogonality graph as CK-31 is unitarily equivalent to it. Any sub-31 KS set must therefore have a different graph. Combined with our result in [20] that all tested algebraic constructions achieving 31 rays produce the same graph as CK-31, this implies that no tested construction yields an alternative 31-vertex graph that might be perturbed downward. Rigidity and algebraic uniformity together suggest that any sub-31 route would require a genuinely new construction principle.

Cortez–Morales-Reyes $N(S)$ invariant [14]. Cortez, Morales, and Reyes introduced the invariant $N(S) = \text{lcm}\{\|v\|^2 : v \in S\}$ for a KS set S with integer coordinates, and proved that S must be defined over the ring $\mathbb{Z}[1/N(S)]$. (Their result is formulated in terms of *algebraic hidden states*—the non-existence of morphisms from partial rings of symmetric matrices—which is a ring-theoretic formulation of contextuality related to but distinct from KS-uncolorability of ray pools.) For CK-31, $N = 30$, corresponding to the ring $\mathbb{Z}[1/30]$. Our Eisenstein pool has $N = 6$, precisely the minimal ring $\mathbb{Z}[1/6]$ predicted by their theorem for Eisenstein-coordinate constructions. This provides an independent algebraic classification: different KS islands inhabit distinct arithmetic strata, and the $N(S)$ invariant constrains which rings need to be searched for each target size.

SI-C closure convergence. Trandafir and Cabello’s 97-ray SI-C closure of CK-31 [19] and our 49-ray integer pool overlap in 37 rays. After a second round of clo-

sure (producing 1741 rays), our entire integer pool is contained in the closure. This convergence of the top-down (closure from CK-31) and bottom-up (alphabet enumeration) approaches suggests that the algebraically accessible landscape of integer-coordinate KS sets is complete, or very nearly so. Rays outside this landscape would require coordinates from genuinely different number fields.

C. Assessment

The two main open problems—improving the lower bound from 24 and proving that 31 is optimal—have very different tractability profiles. Improving the lower bound to 28 or 29 may be achievable with current technology: our finite pool SAT encoding accelerates the LBG pipeline’s realizability bottleneck, and the modulus-2 boundary and plane matching property provide additional constraints that could prune the combinatorial enumeration. Even a modest improvement to 28 or 29 would be significant and publishable. Proving that 31 is optimal is harder. A proof would need to combine geometric realizability obstructions with combinatorial uncolorability in a way that no current technique accomplishes.

Taken together, the results of this series define where a hypothetical sub-31 KS set could hide: it would need to arise from a number field not yet tested, exhibit a novel cancellation identity that our alphabet expansion strategy missed, produce an abstract graph that is both uncolorable and geometrically realizable in $\mathbb{R}^3$, and have no realization in any of the six pools studied here. Each condition is individually possible; their simultaneous satisfaction becomes increasingly implausible in light of the evidence assembled.

XIV. CONCLUSION AND OPEN PROBLEMS

We have described seven complementary strategies—algebraic pool construction and OCUS certification, alphabet expansion, cross-pool mixing, simulated annealing, modulus-based ring analysis, graph perturbation, and structural proof-theoretic analysis—applied systematically to six algebraic ray pools drawn from distinct number rings. In no case was a KS set with fewer than 31 rays found. The central finding is the identification of the *realizability barrier* as the primary obstruction: abstract KS-uncolorable graphs of order 28–30 exist in abundance (61,702 distinct candidates from the perturbation search alone), but $\mathbb{R}^3$ geometry prevents their realization. The difficulty of closing the 24–31 gap is not a shortage of combinatorial candidates but the failure of those candidates to be embeddable in three-dimensional Euclidean space with the required orthogonality structure.

The following problems remain open.

1. **Is 31 optimal?** The central open problem of the field. A proof would require combining geometric

realizability constraints with combinatorial uncolorability in a single argument—a goal that the constraint budget analysis and plane matching results of Section XII approach but do not reach. (*Hard; no current technique is sufficient.*)

2. **Can the lower bound be improved beyond 24?** The finite pool SAT encoding developed here resolves realizability queries in under 1 s that defeat Z3, and is immediately applicable as a screening layer in the LBG combinatorial enumeration pipeline [17]. Geometric constraints from the plane matching property and Trandafir–Cabello rigidity [19] may further prune the search space. An improvement to 28 or 29 may be achievable with current SAT technology combined with these algebraic and geometric constraints. (*Tractable; the most promising near-term target.*)
3. **Does the modulus-2 boundary hold for all number fields?** We have verified that every tested alphabet lacking the coordinate ± 2 fails to produce a KS-uncolorable pool, while every alphabet containing ± 2 succeeds. This has been established for quadratic, cyclotomic, and Pisot algebraic families. Whether the boundary holds for higher-degree algebraic extensions or transcendental coordinates is untested. (*Open; requires new computational experiments or a ring-theoretic argument.*)
4. **Are any of the 394 merged 30-vertex graphs realizable in $\mathbb{C}^3$?** Section XI identifies 394 uncolorable 30-vertex graphs produced by vertex merging, none realizable in $\mathbb{R}^3$ under our tests. Testing realizability over $\mathbb{C}^3$ is a straightforward extension of the numerical optimization method (replacing real dot products with Hermitian inner products) not yet carried out. A positive result would be dramatic: a 30-ray KS set over the complex numbers in dimension three. (*Computable; an accessible near-term experiment.*)
5. **Does the plane matching property yield combinatorial constraints at $n = 30$?** The plane matching observation establishes a structural regularity of all known KS sets at the CK-31 scale. A combinatorial proof that no uncolorable 3-uniform hypergraph with the matching property exists at order 30 would constitute a direct proof of optimality. (*Research question; requires a new combinatorial argument.*)
6. **Can merge saturation be proved for all minimal KS sets?** Vertex merging is verified to preserve KS-uncolorability across all 3,756 tested merges over the six known KS islands, with 100% preservation. This empirical saturation—that every merge of a minimal KS set yields another KS set—suggests a structural characterization of minimal KS sets that, if proved, would substantially

constrain the space of candidates. (*Open conjecture; well-supported computationally.*)

The 24–31 gap remains one of the central open problems in Kochen–Specker theory. The results of this series have mapped the landscape of approaches more completely than previously available and have identified the realizability barrier as the specific geometric obstruction that the search encounters. The evidence assembled here—the algebraic uniformity of all tested constructions, the failure of 61,702 abstract candidates to realize in $\mathbb{R}^3$, the maximally tight constraint budget of CK-31, and the convergence of top-down and bottom-up algebraic explorations—points consistently toward the conjecture that 31 is optimal. A proof remains elusive.

A. Verification methodology

This paper was developed using an LLM-assisted workflow in which all computations, search strategies, and manuscript text were generated by a large language model (Claude, Anthropic) under the author’s direction. To mitigate known failure modes of LLM-generated scientific work—particularly silent fabrication and pressure-induced confabulation [21]—three verification layers were applied: (1) all computational claims are independently reproducible via the scripts in the companion repository

with deterministic seeding (`random.seed(42)`); (2) key results were cross-verified using a competing LLM (GPT, OpenAI) through structured adversarial peer review; (3) all claims about published results were verified by direct reading of the cited papers. All SAT-based claims (uncolorability, OCUS certification, criticality) are machine-checkable proofs within the PySAT framework.

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- [1] S. Kochen and E. P. Specker, “The problem of hidden variables in quantum mechanics,” *J. Math. Mech.* **17**, 59–87 (1967).
 - [2] A. Peres, “Two simple proofs of the Kochen–Specker theorem,” *J. Phys. A: Math. Gen.* **24**, L175–L178 (1991).
 - [3] A. Peres, *Quantum Theory: Concepts and Methods* (Kluwer Academic Publishers, 1993).
 - [4] J. H. Conway and S. Kochen, reported in A. Peres, *Quantum Theory: Concepts and Methods* (Kluwer, 1993).
 - [5] M. Kernaghan, “Bell–Kochen–Specker theorem for 20 vectors,” *J. Phys. A: Math. Gen.* **27**, L829–L830 (1994).
 - [6] M. Kernaghan and A. Peres, “Kochen–Specker theorem for eight-dimensional space,” *Phys. Lett. A* **198**, 1–5 (1995).
 - [7] A. Cabello, J. M. Estebarez, and G. García-Alcaine, “Bell–Kochen–Specker theorem: A proof with 18 vectors,” *Phys. Lett. A* **212**, 183–187 (1996).
 - [8] M. Pavicic, J.-P. Merlet, B. McKay, and N. D. Megill, “Kochen–Specker vectors,” *J. Phys. A: Math. Gen.* **38**, 1577–1592 (2005). arXiv:quant-ph/0409014.
 - [9] F. Arends, J. Ouaknine, and C. W. Wampler, “On searching for small Kochen–Specker vector systems,” in *Proceedings of the International Workshop on Graph-Theoretic Concepts in Computer Science (WG 2011)*, LNCS 6986, 23–34 (2011).
 - [10] M. Pavicic, N. D. Megill, B. C. Waegell, and P. K. Aravind, “Automated generation of Kochen–Specker sets,” *Sci. Rep.* **9**, 6765 (2019). arXiv:1905.01009.
 - [11] S. Uijlen and B. Westerbaan, “A Kochen–Specker system has at least 22 vectors,” *New Generation Computing* **34**, 3–23 (2016).
 - [12] G. Audemard and L. Simon, “On the Glucose SAT solver,” *Int. J. Artif. Intell. Tools* **27**, 1840001 (2018).
 - [13] L. de Moura and N. Bjørner, “Z3: An efficient SMT solver,” in *Proceedings of TACAS 2008*, LNCS 4963, 337–340 (2008).
 - [14] I. Cortez, C. Morales, and M. Reyes, “Minimal ring extensions of the integers exhibiting Kochen–Specker contextuality,” arXiv:2211.13216v4 [math.NT] (2025, preprint).
 - [15] C. Budroni, A. Cabello, O. Gühne, M. Kleinmann, and J.-Å. Larsson, “Kochen–Specker contextuality,” *Rev. Mod. Phys.* **94**, 045007 (2022).
 - [16] M. Kirchweger, A. Peitl, and S. Szeider, “Co-certificate learning with SAT modulo symmetries,” in *Proceedings of AAAI-23*, 4119–4126 (2023).
 - [17] Z. Li, C. Bright, and V. Ganesh, “Leveraging SAT solvers for the minimum Kochen–Specker problem,” in *Proceedings of IJCAI-24*, 1944–1952 (2024). arXiv:2306.13319.
 - [18] A. Cabello, “Simplest Kochen–Specker set,” *Phys. Rev. Lett.* (in press, 2025). arXiv:2508.07335.
 - [19] S. Trandafir and A. Cabello, “Two fundamental solutions to the rigid Kochen–Specker set problem and the solution to the minimal Kochen–Specker set problem under one assumption,” *Phys. Rev. A* **111**, 052204 (2025). arXiv:2501.11640.

- [20] M. Kernaghan, “The algebraic landscape of Kochen–Specker sets in dimension three,” arXiv:XXXX.XXXXX [quant-ph] (2026).
- [21] M. D. Schwartz, “Vibe physics: the AI grad student,” Anthropic Research (2026). Available at <https://www.anthropic.com/research/vibe-physics>.